

CLEAR-NL: Language Complexity Feedback for Accessible Skills Languages

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Skills-based hiring is often presented as a more inclusive alternative to credential-based recruitment, particularly for people whose competences are not well captured by formal qualifications. Central to such approaches are skills languages: standardized descriptions of skills that enable person-job matching and transferability of workers across occupations. However, when skill descriptions are written in abstract or complex language, they can themselves become a barrier, undermining the inclusive aims of skills-based labour market initiatives. In this paper, we conceptualise language complexity in skill descriptions as a fairness and accessibility issue and propose CLEAR-NL: a Dutch-specific, modular approach to support expert maintainers in writing skills in a more accessible way. CLEAR-NL is designed based on four criteria, ensuring that the tool 1) fits in the application context, 2) gives the user feedback that enables informed action, 3) ensures the user is in charge, and 4) is proportional in terms of the size, opacity and risks. We validate CLEAR-NL in a user study with expert maintainers of a skills language, which investigates whether this feedback supports writing and revision decisions. Our findings show that modular feedback can support experts while also highlighting tensions between the design criteria and the practical application context. We discuss the implications of these findings for the design of responsible AI systems that support, rather than replace, human decision-making in fairness-sensitive domains.

Keywords: human-in-the-loop, readability, text complexity, Dutch NLP, skills-based labour market, applied responsible AI

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1 Introduction

Skills-based hiring has gained momentum as a fairer alternative to credential-based recruitment. By shifting the focus from diplomas to competences, such approaches aim to improve labour market mobility and create

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opportunities for people whose skills are not well captured by diplomas, such as people without formal educational credentials, those who have acquired skills through informal career paths, or those returning to the labour market after a prolonged absence [27] [41].

Skills-based approaches rely on unified textual descriptions of skills, formalized in a skills language. These descriptions are developed by domain experts to precisely capture the meaning of a skill. However, skill descriptions are often abstract, dense and linguistically complex, making them hard to understand, especially for people with lower literacy levels [41]. Many people in the Netherlands are in fact weakly literate [54]

Language complexity in skill descriptions therefore risks becoming a barrier to labour market access, rather than a means of inclusion. This makes language complexity a fairness issue.

Existing AI-based approaches to deal with language complexity often fall short in this context. Some methods, such as automatic simplification [25], take over from the user, while others rely on opaque readability scores that provide little insights into why a text is complex [50]. Given the complexity and domain specificity of the context, the expert insights of the people maintaining the skills must be central in formulating the skill descriptions. We thus set out to create a tool to support expert maintainers in reformulating skill descriptions to make them easier to understand. Concretely, we make the following contributions:

- (1) we define criteria to guide the selection and composition of the tool
- (2) we introduce CLEAR-NL: a set of 12 Dutch-specific metrics signalling language complexity that follow from these criteria, and
- (3) we evaluate the applicability and usability of our tool with the end-users of the system (i.e., expert maintainers), validate its design and identify directions for future improvements.

The insights from this work extend beyond skills languages. Many fairness-sensitive applications rely on expert-authored texts, such as public sector communication and policy documents, where linguistic complexity can unintentionally exclude users. The design of CLEAR-NL highlights how language complexity can be framed as a fairness concern, and offers more general lessons for human-in-the-loop AI systems to support expert judgment instead of replacing experts.

We begin by introducing skills languages in more detail in section 2. We discuss their objectives, existing initiatives, and the challenges they raise, zooming in on text complexity as a core accessibility issue. We describe our design criteria and methodology to research suitable methods in section 3, in which we draw on insights from previous readability research. The resulting set of language-complexity metrics is then presented in section 4, and in sections 5 and 6, we report our user study and the feedback on the set of metrics. Finally, in section 7 we discuss the implications of our findings and end with a conclusion in section 8.

2 Background: skills languages

At the core of a skills-based labour market lies a skills language: a standardized way of describing skills that can be shared across sectors and occupations. Skills languages provide a common vocabulary that enables a wide range of downstream applications [27].

For example, a nurse who has been out of the work force for ten years, now wants to return to work but not in a physically demanding role. During his time out of paid employment, he has organized events for a local school. A skills tool helps him identify his existing skills, both from his nursing experience (e.g.: *Nursing*) and from

organizing school events (e.g. *Planning*). Another tool helps him identify relevant jobs, and upskilling opportunities. He finds the job of health care coordinator, for which he possesses most skills, but needs to upskill on managing digital information, to be a better match with the job [42] [41]. See table 1 for some examples of these skill descriptions.

Skill name	Skill description
Nursing	Providing nursing services and medical support to individuals who have medical needs as a result of illness, injury, disability or other health problems.
Planning	Determining in advance what needs to be done, how it needs to be done, who will do it, and what the preconditions are (financially and organizationally) to efficiently achieve a common goal.
Managing (digital) information	Retrieving, storing, and organizing data, information, and content in digital environments and processing them in a structured environment.

Table 1. Examples of skill descriptions in Dutch skills language CompetentNL

Several large-scale skills languages have been developed internationally, such as O*NET [52] in the United States, ESCO [20] in Europe and Competent in Belgium [56][59]. CompetentNL is the Dutch national skills standard and describes three types of skills[13]: human capabilities¹, knowledge (areas)², and language proficiency³[13]. See table 1 for some examples. CompetentNL was developed by multiple Dutch public organizations and will be maintained by experts, responsible for identifying changing or emerging skills, and formulating them to be incorporated into CompetentNL.

Despite the inclusive promise of a skills-based labour market: opening up access for people without formal qualifications, such approaches can still give rise to new forms of exclusion. Skills-based applications can be complicated and abstract because of the "great variety of skills, the context dependency and the linguistic nature of skills tools"[41]. Moreover, the descriptions of the skills themselves are often written in abstract and complicated terms .

In some skills-based applications, the skills defined in the languages require rephrasing to make them accessible to the users [41]. Passport4Work, a skills-based application simplified the labels and descriptions in O*NET to be more accessible for people with lower levels of education. Adapting a skills language for people with lower language skills is a complex and resource-intensive process. This calls for structured approaches that integrate easy-language formulations into skills languages themselves. We thus design a tool to support expert maintainers of CompetentNL in formulating skill descriptions in simpler terms, enabling broader access to skills-based approaches.

3 Design Criteria

We develop a tool that supports the expert maintainers of CompetentNL in formulating skill descriptions in plain language. By effectively assisting their reformulation process, we aim to make the skills language more accessible

¹A type of CompetentNL skill that makes an individual competent to perform a specific task within the context of a profession or function.

²A type of skill from CompetentNL in the form of a clearly defined cluster of content-related, subject-specific facts, principles, theories, and practices that an individual needs to perform a specific task within the context of a profession or function.

³A type of CompetentNL skill that makes an individual competent to perform a specific task within the context of a profession or function.

to people with lower literacy levels. Our aim is not to automate or replace experts, but to provide meaningful support that helps them reflect on and improve the descriptions. We aim to do so in a careful and responsible way. We formulate principles for our tool to inform our choices about the AI and NLP techniques we employ. Within the AI community, there is an ongoing debate about the growing reliance on ever larger language models. While such models can achieve impressive performance, they are also resource-intensive and raise concerns regarding environmental costs, dependence on Big Tech, transparency, and erosion of critical human judgment [5, 57]. These considerations are particularly salient in sensitive application domains such as discrimination detection [57] and education [31], where the societal consequences of automated decisions can be substantial. In such contexts, the benefits of large off-the-shelf models do not automatically outweigh their risks. Ethical proportionality should thus be carefully assessed.

[25] analyse the ethics of text simplification with large language models (LLMs) and highlight both its potential benefits to improving access to information, and its risks, including information loss, introducing incorrect or biased content, and so-called “responsibility gap”: where responsibility for rewritten text becomes unclear. The authors argue that such systems should be designed in close collaboration with domain experts and intended users, ensuring that human experts retain agency.

[16] argue that AI outputs are most useful when they are “actionable”, that is, when a metric enables a user to take informed action. They formulate recommendations for actionable metrics, such as involving the target audience, specificity about the issue it addresses, and clearly defined ideal scores. We adopt actionability as a criterion for our tool.

A further structural issue in NLP research is the strong dominance of English-centric methods and resources. Even multilingual models are often trained primarily on English data, e.g. 93% in the case of GPT-3 [8]. This imbalance has been widely criticised, including through the “Bender rule”, which cautions against implicitly equating “natural language” with English [4]. As a result, many NLP methods are less well suited for Dutch in general, and short skill descriptions in particular. This reinforces the need for language- and context-aware design choices rather than the uncritical application of general-purpose models, ensuring a contextual fit.

Against this backdrop, we formulate the following four selection criteria that guide the design of our tool:

- (1) **Contextual Fit:** The tool must be effective for the specific input of CompetentNL: short Dutch sentences.
- (2) **Actionability:** The tool must give the users feedback that enables them to take an informed action.
- (3) **Agency:** The user must remain responsible for interpreting the skill and be in control of the final decision.
- (4) **Ethical proportionality:** The complexity, opacity, and risks of the tool must be justified given the task, domain, and potential societal consequences.

From the selection criteria it follows that the tool will be modular and transparent. Modularity supports actionability and agency by allowing users to identify which aspects of a skill description contribute to complexity and decide how to act on this feedback. Transparency follows from ethical proportionality. Given the use of short texts, expert users, and high sensitivity to meaning preservation, the risks of complex opaque approaches do not outweigh their benefits.

Based on these criteria, we compiled a list of 49 candidate metrics, from both the scientific literature (among others: [9, 17]), and existing readability implementations [37, 46, 53, 58]. Their applicability was assessed by one

researcher and checked by another (see Appendix A). Twelve metrics were selected for inclusion and are described in Section 4. Below, we outline how the selection criteria filtered out other approaches.

(1) **Contextual Fit**

In the field of automatic readability assessment, readability formulas are prominent. Since the 1940s, over 200 readability formulas have been constructed [44], typically based on word and sentence length. These formulas are usually calibrated for English, and are thus not directly applicable to Dutch. We do select Dutch-specific formulas, such as Flesch-Douma [19].

Readability literature spans a wide range of domains and texts. Many metrics are made for long texts, like paragraphs, articles or books and rely on features such as paragraph length [36]. Frequently occurring metrics therefore focus on features that are not present in the short skill descriptions, such as the number of sentences per paragraph, average sentence length, number of subordinate clauses and clause length [55]. These are very informative indications of complex texts, but due to the short length, skill descriptions do not have this type of complexity.

(2) **Actionability**

Several approaches provide only a single aggregate score or label for text complexity or readability, such as classifier-based readability predictors [38]. Because these metrics do not provide feedback that users can meaningfully interpret or relate to specific revision decisions, they were excluded on the basis that they offer limited support for informed action.

(3) **Agency**

Approaches that automatically rewrite or simplify text, including rule-based and neural simplification systems, were excluded because they shift key decisions away from the expert maintainer. In such systems, choices about wording, and potential loss of meaning are made by the system rather than by the human expert [25].

(4) **Ethical Proportionality**

Finally, some metrics were excluded because their complexity, opacity, or risk profile was disproportionate to the task at hand. In particular, large proprietary language models and opaque neural systems introduce concerns related to environmental cost, vendor dependence, lack of transparency, and responsibility gaps [25, 57]. Given that the task involves short texts, expert users, and high sensitivity to meaning preservation, the added risks of such approaches do not outweigh their potential benefits. The sensitive application domain, assisting people with lower literacy, also demands a careful design. We therefore restrict ourselves to lightweight, transparent metrics.

4 CLEAR-NL: A Modular Tool for Dutch Language Complexity

We present CLEAR-NL: Complex Language identification for Expert Accessibility Revision, a tool comprised of twelve metrics that estimate Dutch short sentence complexity. CLEAR-NL takes as input a skill description and outputs a cross or a tick to indicate if the input is complex or not, an explanation of the cross/tick and explanation of the metric. Figure 1 shows an example. We distinguish three types of metrics: word-level complexity metrics, readability formulas and sentence-level complexity metrics. The latter two also output the scores and thresholds that determine the tick and cross. Below, we discuss the selected metrics and their implementations.

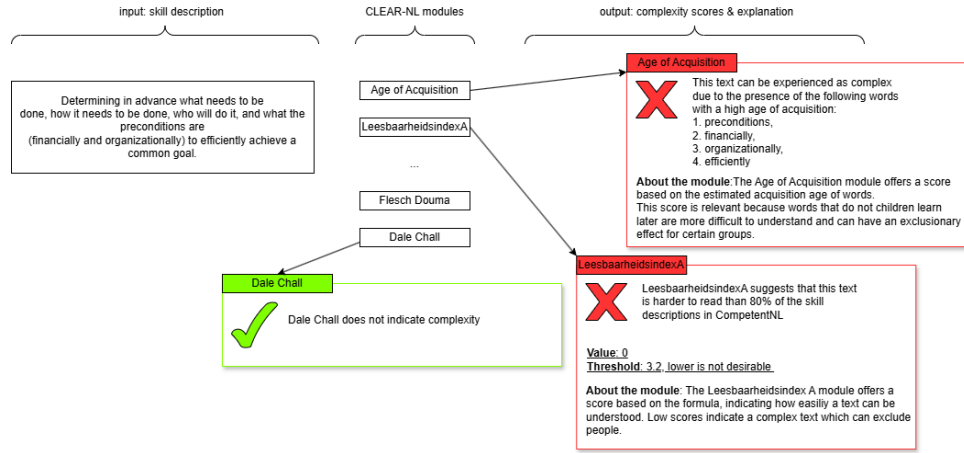


Fig. 1. Schematic view of CLEAR-NL

Metric	Complexity Threshold	Operationalisation
Low Frequency	≥ 1 words Zipf < 3	Dataset SUBTLEX-NL [30]
Low Concreteness	≥ 1 words Concreteness < 3	Dataset: Norms of 3000... [9]
High Age of Acquisition	≥ 1 with AoA ≥ 9	Dataset: Norms of 3000... [9]
Complex Word	≥ 1 words on list	List of 726 complex words [1]
Dale–Chall	> 6.72	$0.1579 \left(\frac{\text{complex_w}}{w} \times 100 \right) + 0.0496 \left(\frac{w}{sen} \right)$
Leesindex A	< 3.2	$195 - 2 \times s_len - \frac{200}{3} \times w_len$
CLIB	< 1	$46 + 0.74 \times freq77 - 6.603 \times w_len - 0.364 \times TTR + 1.425 \times psw$
Flesch–Douma	< 7.1	$206.84 - 0.93 \times \left(\frac{w}{sen} \right) - 77 \times \left(\frac{syl}{w} \right)$
Tuldava	> 6.5	$\frac{syl}{w} \times \ln \left(\frac{w}{sen} \right)$
Sentence length	> 20 words	average number of words per sentence [46]
Monosyllabic fraction	< 0.33	percentage of words with one syllable [12]
Punctuation count	> 3 punctuation marks	number punctuation marks [33]

Table 2. Overview of complexity and readability metrics used in CLEAR-NL.

w = number of words, sen = number of sentences, syl = number of syllables, w_len = average word length, s_len = average sentence length, $freq77$ = percentage of words with cumulative frequency of 77%, TTR : type-token-ratio, psw : percentage of sentences per word

4.1 Word-level text complexity metrics

We select four modules to detect word-level complexity. Each metric is made to detect if the skill description contains a specific type of complex word. In case at least one of such words is found, the sentence itself receives a cross.

(1) Low-frequency words

Words that occur less frequently in everyday language tend to be harder to understand. Therefore, we make use of SUBTLEX-NL, a corpus of Dutch words and their frequencies in everyday language use [30]. We follow the categorisation by the authors, and flag words with $\text{Zipf} < 3$ as infrequent enough to be considered complex. Words absent from the SUBTLEX list are assigned $\text{Zipf} = 1$, i.e. they are treated as very rare and thus more likely to be complex.

(2) Low-concreteness words

Abstract words can be hard to understand. We therefore introduce a metric that quantifies the abstractness of the words in a skill description. For this purpose, we rely on the resource: *Norms of age of acquisition and concreteness for 30,000 Dutch words*. It includes concreteness scores annotated by native Dutch speakers. They were rated on a scale from 1 to 5: 1 - abstract, 2 - more abstract than concrete, 3 - equally abstract and concrete, 4 - more concrete than abstract, 5 - concrete [9]. We label words with concreteness < 3 as complex.

(3) High age-of-acquisition words

Words learned later in life tend to be more complex. We use the *Norms of age-of-acquisition dataset* for the annotations of this property too [9]. The authors point out that the precise values do not represent the actual ages words were learned, but the trend is representative. There is no mapping from age to complexity labels, so we approximate it ourselves. We consider complex words to be words more than 2 standard deviations from the mean. That is, words with an age-of-acquisition of ≥ 9 are labelled as complex.

(4) Complex-word list

We use a curated list of 726 complex words and expressions with suggested alternatives, which was composed by experts in accessible writing for the municipality of Amsterdam [1]. Words or multi-word expressions in skill descriptions that appear on the list are labelled as complex. Alternative formulations are also provided. As the list also indicates the type of problem the flagged item has (either too complex, old-fashioned/archaic/(out)dated, ugly, or legal jargon), this information is also included in the feedback to the user.

4.2 Readability formulas

We select five readability formulas for CLEAR-NL. They output a score based on the features of the skill description. This does not directly translate to a judgment of complexity, so we need to set complexity thresholds. In order to do so, we use a heuristic and calibrate thresholds relative to a reference distribution of existing skill descriptions in CompetentNL. These are very similar to the input CLEAR-NL will be used for, so they are a good reference point. We choose a threshold of 80%: if a skill description is more complex than 80% of the reference skill descriptions for a given metric, it receives a cross. We chose this threshold to focus the experts' attention on only the most clearly complex descriptions, so that alert fatigue is avoided.

(5) Dale–Chall [14]

The Dale–Chall readability score combines sentence length with the proportion of complex words in the sentence. Complex words are those not present on the Dutch basic-word list `basicwords_nl`, containing 3000 most frequent word tokens from the SONAR-500 corpus.

(6) **Leesindex A** [7]

Leesindex A (En: Reading Index A) was developed for Dutch primary school. It gives an indication of complexity by combining the average number of syllables per word and average sentence length.

(7) **CLIB** [47]

CLIB is a Dutch text comprehensibility measure. It quantifies complexity with features not seen in other formulas, such as how often each word is used (type/token ratio) and the percentage of sentences per word.

(8) **Flesch–Douma** [19]

Flesch–Douma is the Dutch adaptation of the well-known Flesch Reading Ease. Its coefficients are tuned for Dutch, but the components remain the same: the number of words per sentence and the syllables per word.

(9) **Tuldava** [51]

Tuldava is a unique formula, because it is language-agnostic: it is based on the assumption that more syllables in a word make it complex, and more words in a sentence make a sentence complex.

4.3 Sentence-level complexity metrics

In addition to the readability formulas, we also select separate linguistic features.

(10) **Sentence length**

Sentence length was found to be one of the most predictive features for complexity prediction in T-Scan [46]. It is also part of some formulas, but we separately include it for its clarity and informativity. [55].

(11) **Fraction of one-syllable words**

Fraction of one-syllable words is a feature in the Coleman-Liau formula [12], based on the idea that short words tend to be easier than long words. This is a clear and informative feature that was not yet captured by the other modules in CLEAR-NL.

(12) **Punctuation count**

Sentences with more punctuation are often structurally more complex. This is a feature in the TRI formula, created for German. We include this feature separately, because punctuation was not captured by the other metrics yet.

4.4 CLEAR-NL Implementation

The twelve metrics and their output form the back-end of CLEAR-NL. The back-end was written in Python, and shared with the end-users in a Docker container to ensure it can be run and integrated into the systems at the end user. To evaluate the usability and configuration of CLEAR-NL, we also developed a user interface. The interface allows users to write or paste a text in the text box, press enter, and receive an analysis of the findings by CLEAR-NL. The results of each metric are visualized by red crosses to indicate complexity and green ticks to indicate no complexity. Along with that, it shows an explanation for each of the metrics. A screenshot of the Dutch UI can be found in Appendix B.

5 User Study

We centre the needs of the end users and therefore performed a user study with the expert maintainers who will use CLEAR-NL to evaluate their experience. We define four research questions:

- (1) What do expert maintainers think of the metrics? What would they like to retain, and what suggestions for improvement do they offer?
- (2) What do expert maintainers think of the user-friendliness of CLEAR-NL? What would they like to retain, and what suggestions for improvement do they offer?
- (3) Does CLEAR-NL provide expert maintainers with sufficient tools to independently make decisions regarding the adjustment of skill descriptions?
- (4) How will expert maintainers use CLEAR-NL in practice?

5.1 Procedure

5.1.1 Sample

CLEAR-NL was tested with five intended users of the module. The five participants are (very) well acquainted with CompetentNL. Four of them indicated that they are closely involved with CompetentNL, both in the development of the ontology and skill descriptions, as well as in communication regarding these matters.

All participants are aware of the importance of effectively conveying information to specific target groups, for example, by writing at B1 level. However, some participants consider themselves more capable of doing so than others, varying from ‘a little’ to ‘(very) good’.

5.1.2 Research design

The study took place in December 2025, at the participants’ workplace. For the research, a private room with a screen was made available. The experiment was conducted by one researcher. CLEAR-NL was run stand-alone on the researcher’s laptop and could be used independently by the participants. Participation was on an individual basis. Each session lasted one hour per person. After a brief introduction by the researcher and the signing of the informed consent by the participant, the participant opened a Word document containing eight skill descriptions to be assessed using CLEAR-NL. It involved four proficiencies and four knowledge areas. Every participant was presented with their own unique set of skills to assess.

The Word document was specially created for this research. In practice, users will receive the skill descriptions to be assessed in various ways (e.g. via e-mail, Excel, Word).

The participants copied the skill descriptions one by one into CLEAR-NL, after which the scores on different metrics were shown. The participants studied the results for each skill description, thinking aloud as they did so. The researcher took notes of these observations and interviewed the participants about their experiences during and after using CLEAR-NL.

5.2 Measures

5.2.1 Skill descriptions

To compile the skill sets, the Skills Taxonomy and Knowledge Area Taxonomy from CompetentNL, version 1, were used⁴. The skill descriptions were selected by the researcher. No concrete selection criteria were applied, except that for each set of skills and knowledge areas, a mix of skills varying in length and complexity was sought.

⁴<https://competentnl.nl/file/download/b2c64c95-4d6e-48a2-84a3-014fbd4ec9b6/documentatie-vaardighedentaxonomie-in-competentnl-versie-1-september-2025.pdf>

The researcher made the preliminary selection based on personal judgment. The skills were subsequently processed through CLEAR-NL, which confirmed that – besides variation in length – the selected skills in each set also varied in complexity. Furthermore, each skill set included one skill that, according to CLEAR-NL, required no adjustment (meaning that none of the metrics showed a cross mark).

In total, participants studied the metrics of 37 out of 40 skill descriptions. Due to time constraints, the last three skill descriptions in one participant's skill set were skipped. The metrics that most frequently detected complexity, and thus most often showed a cross mark to participants, were 'age of acquisition' (33 times), 'concreteness' (32 times), and 'word frequency' (27 times), followed by the 'complex words list for municipality of Amsterdam' (16 times). The metrics that least often detected complexity were 'fraction of one-syllable words' (5 times) and 'average sentence length' (6 times). This can be explained, as the metrics outside the top four – all of which have a threshold value of 80 percent - are calibrated against the entire CompetentNL skills database. As a result, they only detect complex language use in 20 percent of the skill descriptions.

5.2.2 Interview

In order to systematically map participants' experiences with CLEAR-NL, interviews were conducted with participants during and after their use of the module.

After entering each skill description and studying the corresponding metrics, participants were asked about their willingness to take action: whether they would rewrite the definition of the relevant skill based on the outcomes from CLEAR-NL.

Once they had entered all the skill descriptions and reviewed the metrics, participants were interviewed regarding their experiences with the metrics; the user-friendliness of the module; and their knowledge of CompetentNL, communication strategies (ways to effectively convey information to specific target groups), and writing at B1 level.

The questions about the metrics related to the presentation of the metrics (the order of the metrics, the need for (additional) explanation and text suggestions for the metrics); how the metrics are calibrated (too lenient, too strict); and which metrics are considered most valuable, which could be omitted, and which are missing. The questions concerning user-friendliness addressed the module's interface (experiences, wishes) and the tools CLEAR-NL offers for making one's own choices.

5.3 Data analysis

The data were analyzed by the researcher. The main recurring points from the interviews were compiled into a summary report, which was used to answer the research questions. A second TNO researcher checked whether this provided an accurate reflection of the data.

6 Results

6.1 What do expert maintainers think of the metrics? What would they like to retain, and what suggestions for improvement do they offer?

All participants showed interest in the metrics. They analyzed them thoroughly and critically. The reactions to the metrics are mixed, ranging from some cynicism to great enthusiasm. The majority of participants generally

recognize themselves in the outcomes of the metrics. When there is no recognition, participants tend to be more surprised by the metrics with a tick than by those with a cross. When asked, participants would not quickly change the cut-off scores used in the metrics; if anything, they would prefer to relax them further. This is partly because most participants do not always see opportunities to simplify words and sentences. Some participants also believe that certain words and sentences do not necessarily need to be adjusted when they are specific to the field.

The alternative words offered in the ‘complex words list municipality of Amsterdam’ are appreciated by the participants. They find it helpful that suggestions are provided on how to simplify words. They would also like to see (AI-generated) alternatives for complex words and sentences with the other metrics. At the same time, there is also criticism. The alternative suggestions in the ‘complex words list municipality of Amsterdam’ are not always suitable. The metric provides an alternative for a specific word, but does not take into account the context in which it is used. This criticism also applies to the other metrics. Several participants note that the metrics only look at words and sentences (e.g., length), but do not check whether the structure of the sentence is logical and whether the message comes across.

Furthermore, participants do not always understand what a metric measures (e.g., ‘age of acquisition’ and ‘concreteness’) or how the results should be interpreted (e.g., ‘Dale-Chall’, ‘Flesch-Douma’, ‘Tuldava’, and ‘Leesindex A’). This leads to uncertainty about why certain words are labelled as complex. Participants also miss an explanation of what is not correct, making it complex to act upon a metric’s indication of complex language use.

6.1.1 Suggestions

It is therefore important for all participants that an explanation is given for the metrics: what is the metric, what is it based on, and how are the scores determined? This could, for instance, be provided with an information button. Additionally, it would help if the metrics were given a title that clearly reflects their content, which is currently not always the case (e.g., Flesch-Douma). Participants also believe that the metrics should more clearly indicate what causes the complexity (what is not correct), especially for users who do not yet have sufficient experience with the metrics. This applies in particular to the metrics ‘Dale-Chall’, ‘Flesch-Douma’, ‘Tuldava’, and ‘Leesindex A’. On the other hand, the current mention of threshold values and whether the score on the metric needs to go up or down to simplify the skill description seems to add less value for participants.

Two participants question the functionality of the metric ‘fraction of monosyllabic words’. In their view, when using simple language, it is not important to have as many words with a single syllable as possible, but rather to avoid using excessively long words. Simple words can certainly have more than one syllable. If you focus on words with a single syllable, you quickly achieve ‘success’ with a sentence containing many articles, prepositions, and conjunctions.

Furthermore, several participants noticed overlap between the 12 metrics, for example with the 80-percent metrics. This raises the question of whether so many metrics are actually needed, and whether it would not be better to remove or combine some metrics into a single overall score. However, not everyone finds this overlap bothersome. One participant sees overlap between the metrics ‘concreteness’, ‘age of acquisition’, and ‘word frequency’, but believes that this ‘trinity’ in particular has a reinforcing effect.

All things considered, participants find the metrics ‘complex words list municipality of Amsterdam’, ‘average sentence length’, and ‘word frequency’ to be the most valuable. ‘Dale-Chall’, ‘Flesch-Douma’, and ‘Tuldava’ are

most often mentioned as metrics that could be omitted. Two participants would like to see a specific metric that assesses the B1 level of words and sentences.

6.2 What do expert maintainers think of the user-friendliness of CLEAR-NL? What would they like to retain, and what suggestions for improvement do they offer?

There were no critical comments regarding the text box where the skill descriptions are entered, nor the button used to start the evaluation of the description. One participant indicated that they found this part of CLEAR-NL perfectly fine as it is. The other four participants did not say anything about this, but appear to be satisfied with the functionality. In any case, its use does not lead to any problems.

The participants were, however, critical about the presentation of the metrics. They unanimously agreed that it lacks clarity. There is a lot of information, and not everything is visible on one screen, requiring considerable scrolling. One solution is to reduce the size of the screen, but this makes it nearly illegible. Several improvement suggestions were made. For example, two participants proposed presenting the metrics with crosses and ticks side by side rather than underneath each other. Other participants suggested compressing the information (e.g., by combining the 80 percent metrics into a single metric or a single final judgement/score, or by not repeating explanations extensively for metrics that are essentially the same); placing the most essential information for a metric at the front (e.g., by first mentioning the words that are not concrete enough for the 'concreteness' metric, followed by the explanation); adding a filter option so users can determine what they want to see; or freezing the text box with the skill description (as with the first row in Excel).

What most participants would like to retain in the presentation of the metrics is that the metrics with poor scores (the 'crosses') are listed first – in order from worst score to least poor score – with the metrics that score well (the 'ticks') listed underneath or alongside. They believe the order should remain this way, as it allows one to immediately see the 'pain points'.

6.3 Does CLEAR-NL provide expert maintainers with sufficient tools to independently make decisions regarding the adjustment of skill descriptions?

For two participants, CLEAR-NL provides useful tools, but at the same time they indicate that they would also manage without the metrics. They do not need the metrics to determine where the complexity lies and what adjustments are necessary. For the other three participants, CLEAR-NL does not provide sufficient tools to make decisions independently. They need more support.

Participants mainly need additional suggestions (alternatives) for adapting words and sentences. In addition, one participant mentions being helped by a conclusion based on the metrics and an actionable perspective. This participant finds it complex to weigh up all the metrics against each other and make choices. For example, which metric should be given more weight if the outcomes of the metrics are not compatible (e.g., when 'Leesindex A' indicates that the sentence is not easily readable and 'word frequency' shows that the words are not infrequent)? CLEAR-NL does not provide any guidance in this regard.

6.4 How will expert maintainers use the CLEAR-NL in practice?

Generally, participants are more inclined to rely on their own knowledge and insights when rewriting the skill descriptions than to make use of CLEAR-NL. Two participants indicate that they will use the module. The other three participants use the module at most as a confirmation of their own judgement. If they use the metrics at all, it is more likely to be for verification rather than as a reference point. This does not make CLEAR-NL any less valuable, but its application is different from what was originally intended. Since the presumed use of CLEAR-NL is based on a test version, it is possible that a revised version of the module would lead to different usage.

A major challenge for all participants – whether or not they intend to use CLEAR-NL – is the actual rewriting of the skill descriptions. For one participant, it is mainly time-consuming, but for the other four participants, it is also complex in terms of content. They struggle with the question of how a particular sentence or word can be rewritten more simply (at B1 level) without losing the original meaning. The meaning of a word or sentence is often complex to capture using simpler language. CLEAR-NL does not yet provide support in this area. However, there is a clear need for it. To increase its added value for potential users, CLEAR-NL should therefore offer an (AI-generated) alternative for skill descriptions that goes beyond merely providing synonyms and ensures that the meaning of the sentence (or the meaning of the word in the context of the sentence) remains intact.

Finally, we observe that participants struggle with the following questions: For whom am I rewriting the skill descriptions? Should they be written at B1 level? When is it good enough? Do *all* metrics need to be checked off? It is therefore important to make prior agreements and establish clear guidelines upfront to ensure a shared understanding among all domain experts, even if they make little or no use of the language-complexity module. They need clarity about the minimum standard that should be aimed for.

7 Discussion

In this section, we reflect on the findings of the user study and discuss their implications for the design of CLEAR-NL and for the design of responsible language-support tools in fairness-sensitive domains more broadly. These insights help us understand how expert users interact with metric-based feedback and where tensions arise between contextual fit, actionability, agency and ethical proportionality. One limitation of the study is that it involved a small sample of five participants who were highly experienced and closely involved in the development and maintenance of CompetentNL. Their needs may differ from those of future users with less domain expertise or less experience writing accessible language. Evaluating CLEAR-NL with a broader and more diverse group of users is an important direction for future work.

7.1 Implications and potential improvements for CLEAR-NL

The user study revealed both strengths and limitations of the current design of CLEAR-NL. While participants generally recognised the value of modular, transparent feedback, their experiences highlight several areas where it could be improved to better support expert maintainers in practice.

7.1.1 (AI-generated) alternatives for skill descriptions

A clear finding from the user study is that there is a need for assistance in rewriting the complex skill descriptions. CLEAR-NL currently functions primarily as a signalling and reflection tool: it helps users identify potential sources of linguistic complexity, but offers limited support for rewriting skill descriptions in simpler language.

This limitation is closely tied to the design criteria that guided the development of CLEAR-NL. In prioritising actionability and ethical proportionality, we deliberately avoided automatic rewriting or simplification, as they can introduce a responsibility gap and semantic drift.

Future iterations of CLEAR-NL could therefore explore the careful inclusion of alternative formulations. LLMs are the logical choice for this task, but given our criteria of agency and proportionality, including them in CLEAR-NL requires careful consideration. A factor to consider here is that expert maintainers may already turn to LLMs outside of CLEAR-NL to assist with rewriting, when they struggle to find a good reformulation. From this perspective, excluding generative support from CLEAR-NL does not necessarily prevent its use, but rather shifts it beyond the scope of institutional oversight. This raises the question of whether a carefully constrained integration could offer a more responsible alternative to unregulated, ad-hoc use. Rather than replacing expert judgment, such integration could function as a governed aid. Exploring this tension between exclusion and controlled inclusion remains an important direction for future work.

7.1.2 Reducing information overload

Participants reported that the number of metrics and the way they are presented can be overwhelming. This reflects a structural tension in the modular approach: we decomposed complexity into multiple components to improve interpretability, but simultaneously increase the cognitive load.

From the perspective of actionability, we see that more information does not necessarily translate to better support for decision-making. Actionability depends not only on transparency, but also on the user's ability to prioritise and act on signals.

In future iterations of CLEAR-NL, we can therefore increase actionability by decreasing the amount of metrics, for example by combining all word-level metrics into one "complex word"-metric and adding "not concrete", "not frequent", "high age-of-acquisition" or "on a complex word list" as the "reason for complexity".

7.1.3 More detailed explanations for the metrics

Users also indicated that the output of CLEAR-NL was not always sufficiently clear. Several participants reported that they did not fully understand what some metrics measure or how their scores should be interpreted, especially for abstract readability formulas.

This limitation is closely related to our initial interpretation of agency as leaving the assessment of complexity primarily to the expert user. However, the findings suggest that agency is not only experienced through freedom of interpretation, but also through having access to a complete explanation of the system's reasoning. In future iterations of CLEAR-NL, we can include a more elaborate explanation of the metrics, what they are based on, and how the scores are calculated. Providing such explanations optionally (on demand) would allow users to decide how much support they need, while preserving expert control and avoiding unnecessary information overload.

7.1.4 Clarity about the goal

Finally, participants indicated that CLEAR-NL did not sufficiently support them in determining when a skill description could be considered "good enough." This feedback points primarily to an expectation-setting issue rather than a missing functionality. In line with the design criterion of agency, CLEAR-NL does not aim to determine whether a sentence is correct or optimal, as there is no objectively correct formulation of a skill description. Such

judgments necessarily depend on context, target audience, and expert interpretation, and therefore remain the responsibility of the domain experts themselves.

At the same time, the findings suggest that the absence of an explicit affirmation can be experienced as a lack of support. Future iterations of CLEAR-NL could address this by more clearly communicating the scope and intent of the tool, for example by explaining who are the users of the skill descriptions in CompetentNL, and making explicit that the absence of flagged issues indicates that no potential complexity has been detected, rather than that a sentence meets a definitive standard.

Overall, the findings reveal friction between CLEAR-NL's design criteria and the realities of expert practice, where abstract principles must be negotiated in situated and pragmatic ways. At the same time, the scope of this work is limited to language complexity in skill descriptions. We did not evaluate downstream effects, such as how revisions influence matching algorithms, job recommendations, or labour-market outcomes, nor does improved accessibility address all dimensions of fairness. Even with accessible formulations, biases may persist in how skills are prioritised or operationalised. This positions CLEAR-NL as one element within a broader fairness-aware ecosystem.

8 Conclusion

In this paper, we presented CLEAR-NL, a modular and transparent tool for identifying language complexity in skill descriptions. CLEAR-NL was developed based on four guiding criteria: contextual fit, actionability, agency and ethical proportionality, and operationalised through a modular set of 12 language complexity metrics. Rather than producing a single opaque score or automatically rewriting text, CLEAR-NL provides interpretable signals that support expert reflection and informed decision making.

A qualitative user study with expert maintainers shows that such feedback can support reflection and reformulation on complex language use. At the same time, the findings reveal important tensions. While transparency is important, too much information can overwhelm the user. Similarly, agency should be maintained, but it should not hinder effective action, and while automatic rewriting was intentionally excluded, users still require assistance with reformulation.

These findings underline that modular, explainable metrics are a viable foundation for responsible language complexity tools, but also that their effectiveness depends on thoughtful embedding in practice. More broadly, this work contributes to discussions on responsible AI by illustrating how support tools can be designed to assist rather than replace human experts. Future work may explore how CLEAR-NL can be integrated into workflows, be extended with generative AI support and be evaluated on downstream impacts on the labour market.

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